

Practical 2

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2.1.1 List Operations

Problem Statement:

Write a Python program that implements a menu-driven interface for managing a list of integers. The program should have the following menu options:

1. Add
2. Remove
3. Display
4. Quit

The program should repeatedly prompt the user to enter a choice from the menu.

Depending on the choice selected, the program should perform the following actions:

- **Add:** Prompts the user to enter an integer and add it to the integer list. If the input is not a valid integer, display "Invalid input".

- **Remove:** Prompts the user to enter an integer to remove from the list. If the integer is found in the list, remove it; otherwise, display "Element not found". If the list is empty, display "List is empty".
- **Display:** Displays the current list of integers. If the list is empty, display "List is empty".
- **Quit:** Exits the program.
- The program should handle invalid menu choices by displaying "Invalid choice". Ensure that the program continues to prompt the user until they choose to quit (option 4).

Code:

```
l1 = []
while True:
    print("1. Add") print("2.
Remove") print("3.
Display")      print("4.
Quit")

    n=int(input("Enter choice: ")) if
        n==1:
            add = int(input("Integer: "))
l1.append(add)
    print(f"List after adding: {l1}") elif
n==2: if len(l1)==0:    print("List is
empty")                elif    len(l1)!=0:
remove=int(input("Integer: "))    if
remove not in l1:
                                print("Element not found")
        else:
```

```

        l1.remove(remove)    print(f"List after
removing: {l1}") elif n==3: if len(l1)==0:
        print("List is
empty")    else:
print(l1) elif n==4:
        break
else:
        print("Invalid choice")

```

Output:

Average time: 0.217 s (217.00 ms) Maximum time: 0.309 s (309.00 ms)

2 out of 2 shown test case(s) passed
1 out of 1 hidden test case(s) passed

Test case 1 (202 ms) Debug

Expected output	Actual output
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 1	Enter choice: 1
Integer: 11	Integer: 11
List after adding: [11]	List after adding: [11]
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 1	Enter choice: 1
Integer: 25	Integer: 25
List after adding: [11, 25]	List after adding: [11, 25]
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 2	Enter choice: 2
Integer: 26	Integer: 26
Element not found	Element not found
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit

Average time

0.217 s

217.00 ms

Maximum time

0.309 s

309.00 ms

2 out of 2 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Enter choice: 2	Enter choice: 2
Integer: 11	Integer: 11
List after removing: [25]	List after removing: [25]
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 2	Enter choice: 2
Integer: 25	Integer: 25
List after removing: []	List after removing: []
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 3	Enter choice: 3
List is empty	List is empty
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 2	Enter choice: 2
List is empty	List is empty
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit

Average time

0.217 s

217.00 ms

Maximum time

0.309 s

309.00 ms

2 out of 2 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Enter choice: 8	Enter choice: 8
Invalid choice	Invalid choice
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 4	Enter choice: 4

Test case 2

140 ms

Debug

Expected output	Actual output
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 1	Enter choice: 1
Integer: 1	Integer: 1
List after adding: [1]	List after adding: [1]
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 1	Enter choice: 1
Integer: 2	Integer: 2
List after adding: [1, 2]	List after adding: [1, 2]
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit

Average time 0.217 s 217.00 ms Maximum time 0.309 s 309.00 ms		2 out of 2 shown test case(s) passed 1 out of 1 hidden test case(s) passed	
Enter choice: 1	Enter choice: 1	Enter choice: 1	Enter choice: 1
Integer: 3	Integer: 3	Integer: 3	Integer: 3
List after adding: [1, 2, 3]	List after adding: [1, 2, 3]	List after adding: [1, 2, 3]	List after adding: [1, 2, 3]
1. Add	1. Add	1. Add	1. Add
2. Remove	2. Remove	2. Remove	2. Remove
3. Display	3. Display	3. Display	3. Display
4. Quit	4. Quit	4. Quit	4. Quit
Enter choice: 1	Enter choice: 1	Enter choice: 1	Enter choice: 1
Integer: 4	Integer: 4	Integer: 4	Integer: 4
List after adding: [1, 2, 3, 4]	List after adding: [1, 2, 3, 4]	List after adding: [1, 2, 3, 4]	List after adding: [1, 2, 3, 4]
1. Add	1. Add	1. Add	1. Add
2. Remove	2. Remove	2. Remove	2. Remove
3. Display	3. Display	3. Display	3. Display
4. Quit	4. Quit	4. Quit	4. Quit
Enter choice: 2	Enter choice: 2	Enter choice: 2	Enter choice: 2
Integer: 2	Integer: 2	Integer: 2	Integer: 2
List after removing: [1, 3, 4]	List after removing: [1, 3, 4]	List after removing: [1, 3, 4]	List after removing: [1, 3, 4]
1. Add	1. Add	1. Add	1. Add
2. Remove	2. Remove	2. Remove	2. Remove
3. Display	3. Display	3. Display	3. Display
4. Quit	4. Quit	4. Quit	4. Quit
Enter choice: 2	Enter choice: 2	Enter choice: 2	Enter choice: 2
Integer: 6	Integer: 6	Integer: 6	Integer: 6
Element not found	Element not found	Element not found	Element not found
1. Add	1. Add	1. Add	1. Add
2. Remove	2. Remove	2. Remove	2. Remove
3. Display	3. Display	3. Display	3. Display
4. Quit	4. Quit	4. Quit	4. Quit

Enter choice: 3	Enter choice: 3
[1, 3, 4]	[1, 3, 4]
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 4	Enter choice: 4

2.1.2

DictionaryOperations

Problem Statement:

Write a Python program to perform insertion, update, deletion, and traversal operations on a dictionary. An initial dictionary containing 10 predefined records is already given in the program.

Operations to be Performed:

1. Insertion – Insert a new key-value pair into the dictionary using user input.
2. Update – Update the value of an existing key using user input.
3. Deletion – Delete a specified key from the dictionary using user input.
4. Traversal – Traverse the final dictionary and display all key-value pairs.

Input and Output Format:

1. Read an integer representing the key to be inserted and a string representing its value. Insert this new key-value pair into the dictionary and display the dictionary after insertion.
2. Read an integer representing the key to be updated and a string representing the new value. Update the value of the specified key (only if it exists) and display the dictionary after the update.
3. Read an integer representing the key to be deleted. Delete the specified key from the dictionary (only if it exists) and display the dictionary after deletion.
4. Finally, traverse the dictionary and display all key-value pairs. The program should also display the original dictionary before performing any operations. Each output should be printed with appropriate messages.

Code:

```
# Initial dictionary with 10
predefined records student = {
1: "Amit",
    2: "Riya",
    3: "Kiran",
    4: "Neha",
    5: "Arjun",
    6: "Pooja",
    7: "Rahul",
8: "Sneha",
9:
"Vikram",
10: "Anjali"
}
# write your code here..
print("Original Dictionary:", student)

new_key = int(input())
new_value = input()
student[new_key] = new_value
print("After Insertion:", student)
update_key = int(input())
new_val = input()
if update_key in student:
    student[update_key] =
new_val
print("After Update:",
student)
del_key = int(input())
if del_key in student:
    del student[del_key]
print("After Deletion:",
student)
print("Traversing
Dictionary:")
for key,
```

```
value in student.items():  
    print(f"{key} : {value}")
```

Output:

Average time
0.098 s
98.25 ms

Maximum time
0.139 s
139.00 ms

✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 85 ms Debug

Expected output

Original Dictionary: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali'}

11

Suresh

After Insertion: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}

3

Karthik

After Update: {1: 'Amit', 2: 'Riya', 3: 'Karthik', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}

5

After Deletion: {1: 'Amit', 2: 'Riya', 3: 'Karthik', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}

Traversing Dictionary:

1: Amit

2: Riya

3: Karthik

4: Neha

6: Pooja

7: Rahul

8: Sneha

9: Vikram

10: Anjali

11: Suresh

Actual output

Original Dictionary: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali'}

11

Suresh

After Insertion: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}

3

Karthik

After Update: {1: 'Amit', 2: 'Riya', 3: 'Karthik', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}

5

After Deletion: {1: 'Amit', 2: 'Riya', 3: 'Karthik', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}

Traversing Dictionary:

1: Amit

2: Riya

3: Karthik

4: Neha

6: Pooja

7: Rahul

8: Sneha

9: Vikram

10: Anjali

11: Suresh

Average time:

0.098 s

98.25 ms

Maximum time:

0.139 s

139.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 2

71 ms

Debug

Expected output	Actual output
Original Dictionary: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali'}	Original Dictionary: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali'}
12	12
Meera	Meera
After Insertion: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 12: 'Meera'}	After Insertion: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 12: 'Meera'}
6	6
Divya	Divya
After Update: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Divya', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 12: 'Meera'}	After Update: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Divya', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 12: 'Meera'}
1	1
After Deletion: {2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Divya', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 12: 'Meera'}	After Deletion: {2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Divya', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 12: 'Meera'}
Traversing Dictionary:	Traversing Dictionary:
2: Riya	2: Riya
3: Kiran	3: Kiran
4: Neha	4: Neha
5: Arjun	5: Arjun
6: Divya	6: Divya
7: Rahul	7: Rahul
8: Sneha	8: Sneha
9: Vikram	9: Vikram
10: Anjali	10: Anjali
12: Meera	12: Meera

2.2.1 Linear Search Technique

Problem Statement:

Write a program to check whether the given element is present or not in the array of elements using linear search.

Input format:

- The first line of input contains the array of integers which are separated by space

- The last line of input contains the key element to be searched

Output format:

- If the element is found, print the index.  If the element is not found, print **Not found**.

Code: def

```
linear_search(arr,
key):    for i in
range(len(arr)):
if arr[i] == key:
return i    return
-1
```

```
arr =
list(map(int,input().split())) key
= int(input()) result =
linear_search(arr, key) if
result != -1:
    print(result) else:
    print("Not found")
```

Output:

The screenshot displays the test results for a program. At the top, a summary bar indicates that all tests passed: '2 out of 2 shown test case(s) passed' and '2 out of 2 hidden test case(s) passed'. Performance metrics show an average time of 0.026 s (26.25 ms) and a maximum time of 0.029 s (29.00 ms). Below this, two test cases are detailed. Test case 1, which took 29 ms, shows an expected output of '1 2 3 4 3 5 6', '3', and '2', which matches the actual output. Test case 2, which took 24 ms, shows an expected output of '1 2 3 4 5 6 7 8 9 19', '20', and 'Not - found', which also matches the actual output. Each test case entry includes a 'Debug' button and icons for viewing the code and test cases.

Metric	Value
Average time	0.026 s (26.25 ms)
Maximum time	0.029 s (29.00 ms)
Shown test cases	2 out of 2 passed
Hidden test cases	2 out of 2 passed

Test Case	Expected Output	Actual Output
Test case 1 (29 ms)	1 2 3 4 3 5 6 3 2	1 2 3 4 3 5 6 3 2
Test case 2 (24 ms)	1 2 3 4 5 6 7 8 9 19 20 Not - found	1 2 3 4 5 6 7 8 9 19 20 Not - found

2.2.2 Captain of the Team

Problem Statement:

You are provided with the heights of 11 cricket players (in centimeters). Your task is to identify the tallest player, who will be selected as the captain of the team.

Input Format:

The first line of input will contain 11 integers, each representing the height of a player (in centimeters), each separated by a space.

Output Format

The output should be the height (in centimeters) of the tallest player.

Code:

```
heights = list(map(int,input().split()))
tallest_player = max(heights)
print(tallest_player)
```

Output:

Average time

0.023 s

23.33 ms

Maximum time

0.032 s

32.00 ms

✓ 1 out of 1 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1

22 ms

Debug

Expected output

Actual output

171 169 185 156 174 191 186 190 187 172 160

171 169 185 156 174 191 186 190 187 172 160

191

191